BTC Journal Club: Bulk RNA-Seq Workshop

David de la Cerda

4/03/2026

Outline

Background

- Affymetrix platform → bulk RNA-Seq → scRNA-Seq
- Data types to consider
- Publicly available databases
  - What to look for when re-analyzing data from a publication

Statistical Methodologies

- Model considerations
- Software assumptions

Bioinformatics tools

- Shell/R scripting best practices
- Processing raw FASTQ files to BAM files
- Quantifying BAM files
- Normalizing/Transforming data and statistical Analysis (Hands-on portion)

Early gene expression technologies are several *in situ* experiments on a chip

- Early 2000's offered Microarray technology (ex: Affymetrix or nCounter NanoString)
- Have a set of probes (pre-determined oligo-nucleotides) on a chip
- Extracted/amplified RNA from sample is fluorescently labeled and will hybridize to the nucleotides on the chip
- Laser excitation of the probes will allow them to emit colors (usually three)
- Analysis is performed based on image intensity for a given oligo site

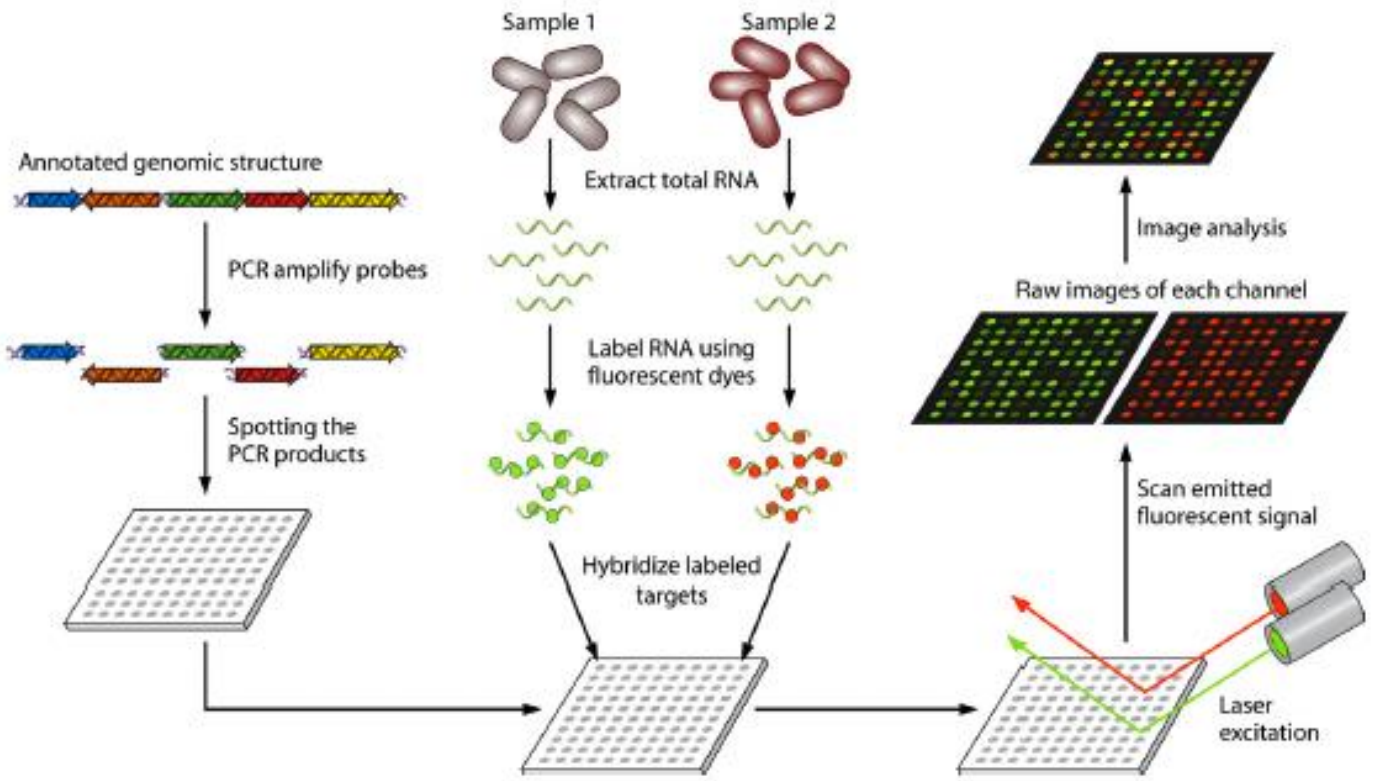


Fig 2. Miller & Tang 2009 *Clin Microbiol Rev*

Microarray experiments have limitations

- Costly at the time and oftentimes customization within core facilities was a better option
- Limited to a set of probes that hybridize to the chip (usually not capturing the full genome or novel genes/isoforms)
- Poor resolution/detection of intensity
- Statistical methods at the time had to incorporate several steps to attempt to capture true positive signals

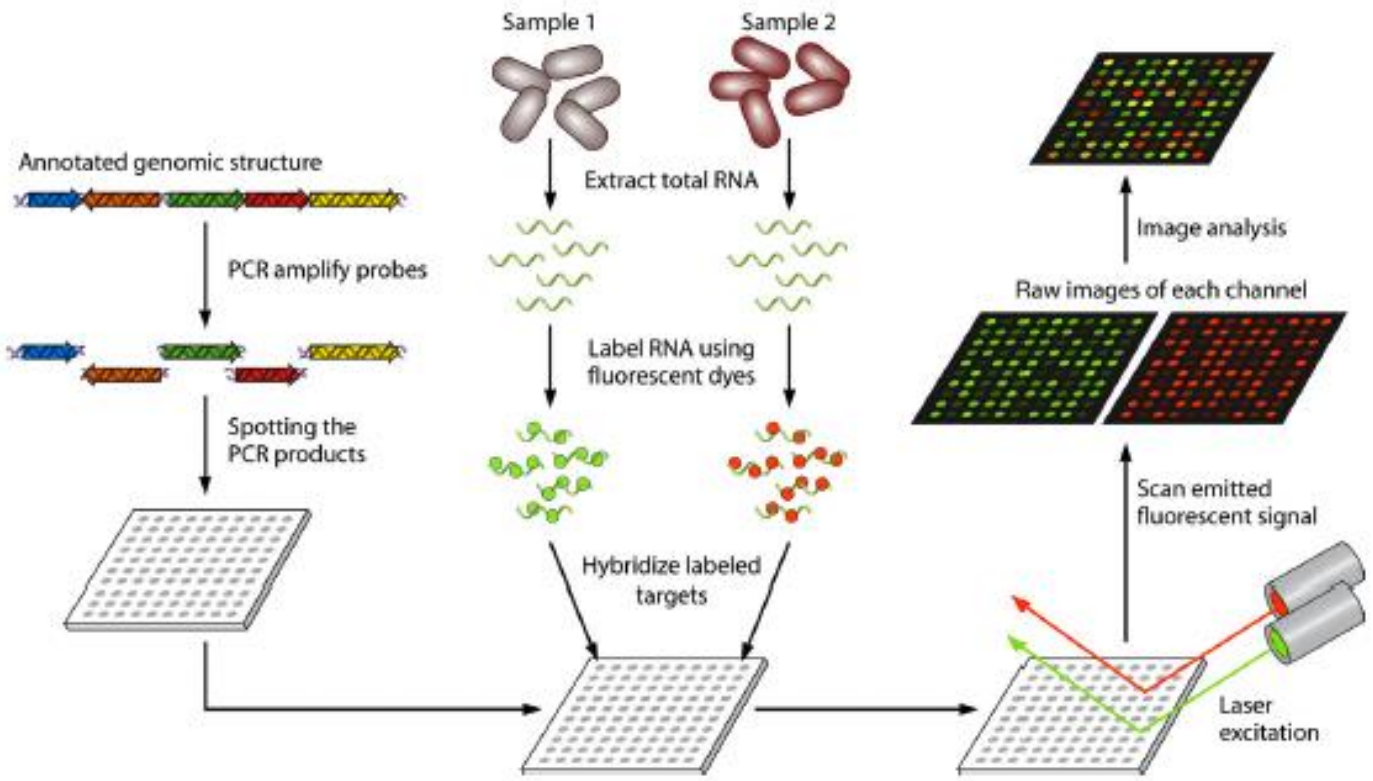
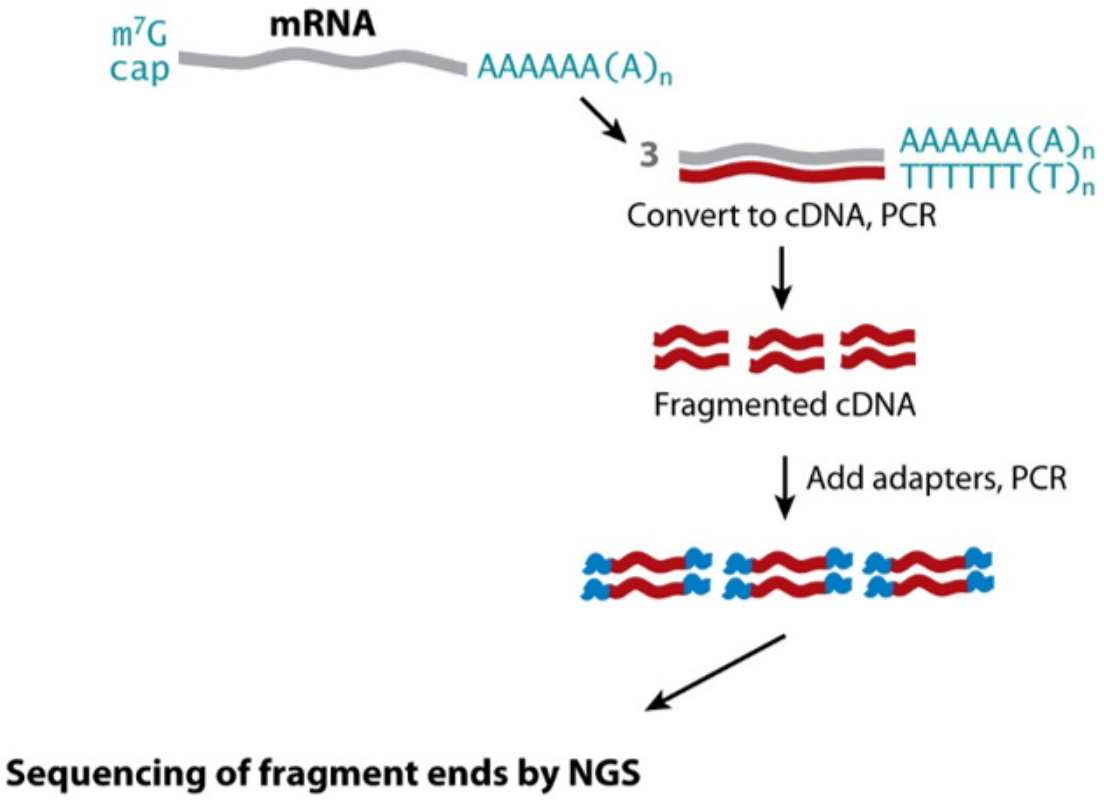


Fig 2. Miller & Tang. 2009 *Clin Microbiol Rev*

Bulk RNA-Seq became a more popular and eventually more affordable option given the rise in NGS technologies

RNA-Seq is used to quantify gene expression genome-wide



Adapted from Simon et al. 2009. *Ann. Rev. Plant Biol.*

Raw Reads → Processed Reads → Alignment to Reference → Fragment Counting → Differential Expression Analysis

Raw NGS data are in the form of FASTA/FASTQ files

- These NGS files are usually quite large and can be a single file or multiple files that refer to a particular sample (ex: paired-end sequencing)
- Basic FASTA file only has header and sequence
- FASTQ has title/header usually has platform information or quality score information

```
>SEQUENCE_1
MTEITAAMVKELRESTGAGMMDCKNALSETNGDFDKAVQLLREKGLGKAAKKADRLAAEG
LVSVKVSDDFTIAAMRPSYLSYEDLDMTFVENEYKALVAELEKENEERRRLKDPNKPEHK
IPQFASRKQLSDAILKEAEEKIKEELKAQGKPEKIWDNIIPGKMNSFIADNSQLDSKLTL
MGQFYVMDDKKTVEQVIAEKEKEFGGKIKIVEFICFEVGEGLEKKTEDFAAEVAAQL
>SEQUENCE_2
SATVSEINSETDFVAKNDQFIALTKDTTAHIQSNSLQSVEELHSSTINGVKFEEYLKSQI
ATIGENLVVRRFATLKAGANGVVNGYIHTNGRVGVVIAAACDSAEVASKSRDLLRQICMH
```

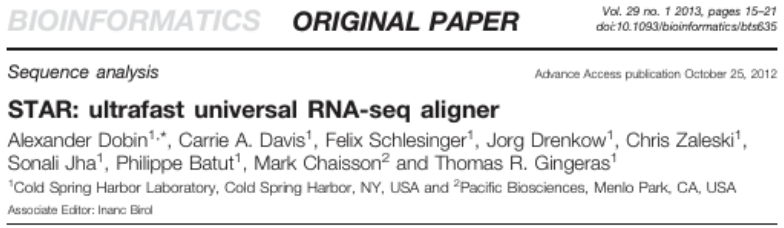
Header Sequence Quality

```
@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979
GGAGGAAGGTCCTCGCTCCTCTTTCATATAAGGGAAATGGCTGAAT
+
FFFFHHHHHHJIJJJJJJJJIJJJIGIGIGGIJJIJIJJJJJJIII
@HWI-ST227:389:C4WA2ACXX:7:1205:15214:42893
GAGGATCCCAGGGAGGAAGGTCCTCGCTCCTCTTTCATCTAAGGGA
+
12BAFB?A:3<AE1@<FF;1*@EG*)?0?DBD>9BF9B*?######
@HWI-ST227:389:C4WA2ACXX:8:2208:2467:44624
AAAGAGGAGAGAGGACCATCCTCCCTGGGATCCTCAGAAGTCTACT
+
BDDA:DB?2AA@FC>F?EEGC<FED>GFD;?GBB?<?F99*/9?9?
```

https://en.wikipedia.org/wiki/FASTA_format

https://biocorecrg.github.io/RNAseq_course_2019/fastq.html

Sequenced short reads are aligned to the reference genome


BIOINFORMATICS ORIGINAL PAPER

Vol. 29 no. 1 2013, pages 15–21
doi:10.1093/bioinformatics/bts635

Sequence analysis

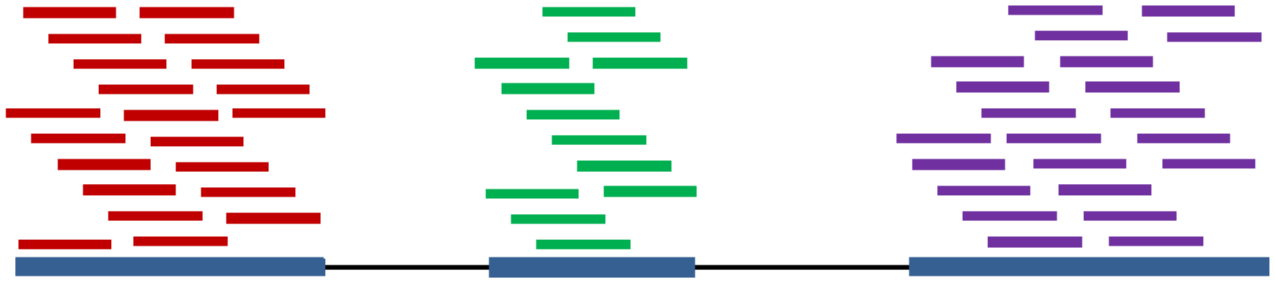
Advance Access publication October 25, 2012

STAR: ultrafast universal RNA-seq aligner

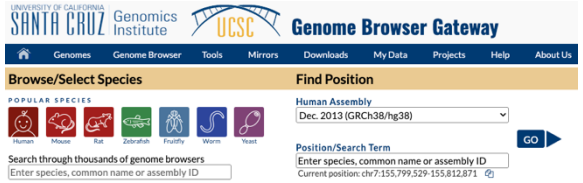
Alexander Dobin[1,*], Carrie A. Davis[1], Felix Schlesinger[1], Jorg Drenkow[1], Chris Zaleski[1], Sonali Jha[1], Philippe Batut[1], Mark Chaisson[2] and Thomas R. Gingeras[1]

[1]Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA and [2]Pacific Biosciences, Menlo Park, CA, USA

Associate Editor: Inanc Birol



GRCh38/hg38



Raw Reads → Processed Reads → Alignment to Reference → Fragment Counting → Differential Expression Analysis

Aligned files are in SAM or BAM format

- Aligned files are in **S**equence **A**ligned **M**ap or **B**inary **A**ligned **M**ap format
    - SAM is a text-based file
    - BAM is indexed and compressed
    - Use the information in these files for quantification or visualizing specific regions of the genome
- CRAM can be used to look at coordinates in software like IGV (Integrated Genome Viewer)

Field	Value
Query name	D00137:453:HLFY2BCXY:2:1115:10428:98737
FLAG	419 *
Reference name	chr10
Leftmost mapping position (1-based)	35442
Mapping quality	3 *(p=0.5)*
CIGAR string	13M174562N32M6S *
Reference sequence name of the primary alignment of the mate	= *same chromosome*
Position of the primary alignment of the mate	236864
observed fragment length	201473
Sequence	AGCTGTTATTGAACAAGAAGGGATTGGTTGCCAGGAGATGAGATTAGCATT
Quality	@DDD?<1D1CGEE11<C@GHIIH?@E?G@CHH? FH@0GH@C«<@1DFCHH

https://biocorecrg.github.io/RNAseq_course_2019/aInpractical.html

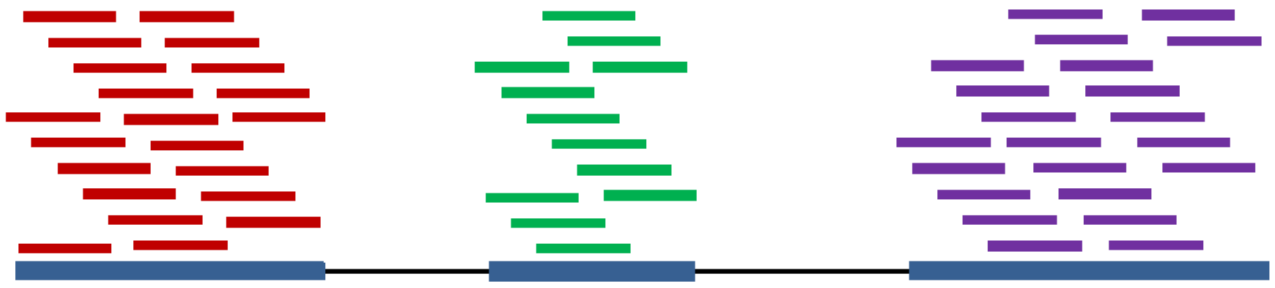
The number of reads mapping to each gene is quantified to estimate expression

JOURNAL ARTICLE

featureCounts: an efficient general purpose program for assigning sequence reads to genomic features FREE

Yang Liao , Gordon K. Smyth , Wei Shi Author Notes

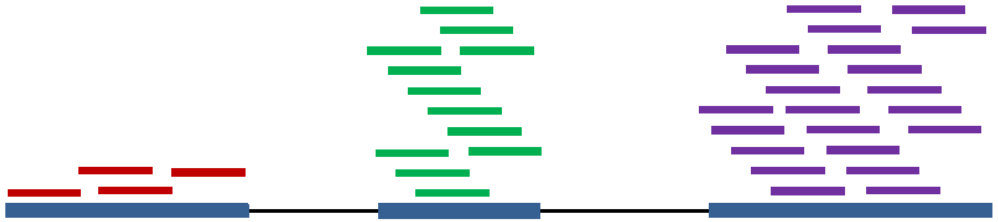
Bioinformatics, Volume 30, Issue 7, April 2014, Pages 923–930,
https://doi.org/10.1093/bioinformatics/btt656
Published: 13 November 2013 **Article history**

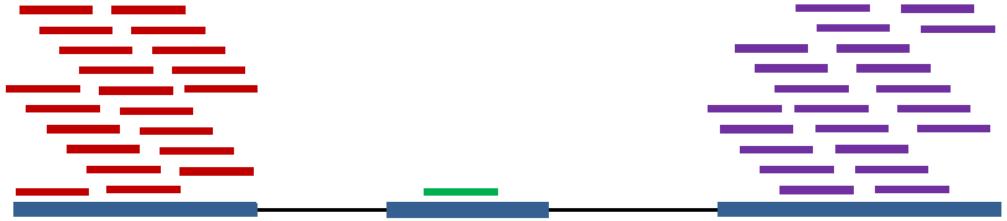


ENSG00000100450 ENSG00000150782 ENSG00000120949

Raw Reads → Processed Reads → Alignment to Reference → Fragment Counting → Differential Expression Analysis

Statistically test for expression changes between wild-type and treated samples





Wild-type sample

Treatment sample

Raw Reads → Processed Reads → Alignment to Reference → Fragment Counting → Differential Expression Analysis

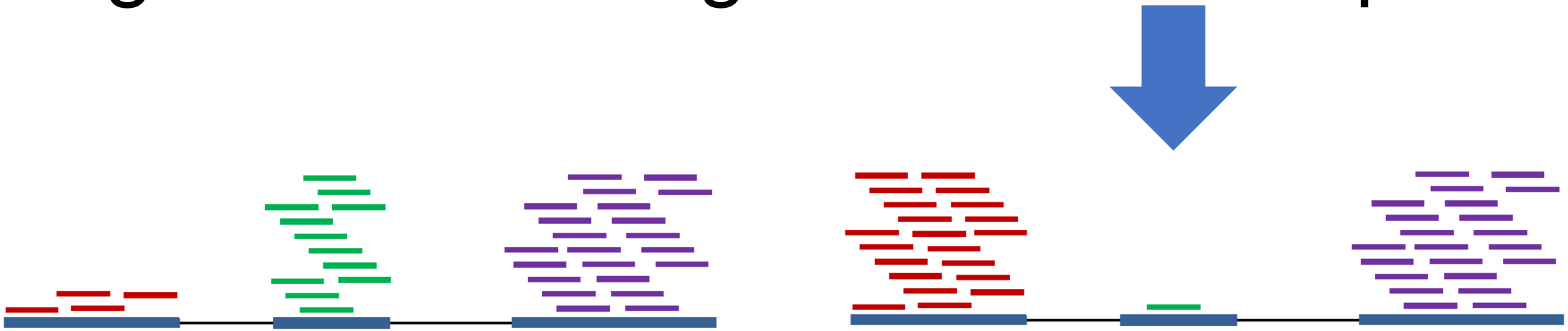
Positive fold change in treated samples

Wild-type sample

Treatment sample

Raw Reads → Processed Reads → Alignment to Reference → Fragment Counting → Differential Expression Analysis

Negative fold change in treated samples



Wild-type sample

Treatment sample

Raw Reads → Processed Reads → Alignment to Reference → Fragment Counting → Differential Expression Analysis

Quantified counts are used to perform differential expression analysis

- Count files will have genes/features as rows and samples as columns
- Genes will typically be in ENSEMBL IDs or Gene Symbols

gene	A549_0_1chr10	A549_0_2chr10	A549_0_3chr10	A549_25_1chr10	A549_25_2chr10	A549_25_3chr10
ENSG00000260370.1	0	0	1	0	1	1
ENSG00000237297.1	10	8	10	12	5	2
ENSG00000261456.5	210	320	291	300	267	222
ENSG00000232420.2	3	2	0	1	2	6

https://biocorecrg.github.io/RNAseq_course_2019/differential_expression.html

Bulk vs single-cell experiment considerations

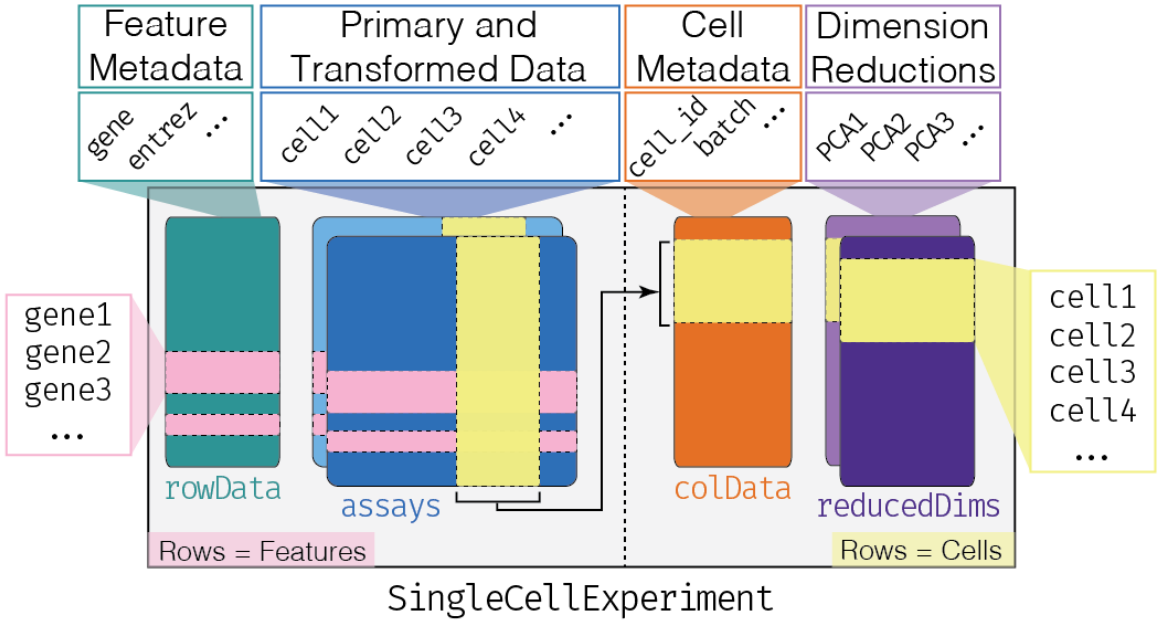
Bulk RNA-Sequencing provides aggregated reads for all cells in a given sample

- Ex: If you provide all lymphocytes of a sample you cannot distinguish which transcripts are coming from specific cell types
- Important to isolate a tissue or cell type **before** sequencing
- These experiments are favorable when you want extensive read coverage for an experiment

Single-cell experiment data are stored to capture cell and gene information

Single-cell RNA-Sequencing provides reads for **specific cells** for a given sample

- Ex: If you provide all lymphocytes of a sample, cell-barcoding and deconvolution methods can help assign which reads are coming from specific cells
- These experiments provide less read coverage, but can be helpful if you have a heterogeneous cell population that you want to annotate



https://alexslemonade.github.io/training-modules/scRNA-seq-advanced/01-read_filter_normalize_scRNA.nb.html

Many studies make their RNA-Seq data publicly available, but there are several considerations to take into account

Before querying a database, look at the methods section of the publication and look for key information:

What kind of data are collected?

- Total RNA – less common in recent studies but will most likely contain all RNA species for a sample

- polyA selected – typically referred to as mature or processed mRNA or the protein-coding fraction of the sample

- Ribo-depleted – Remove most Ribosomal RNAs (these make up more than 80% of the molecules in a total RNA sample) but can better capture non-coding RNA sequences that may be of interest

Zhao et al. 2018. *Scientific Reports*

After assessing the data type, decide if what is available will help you answer your question of interest

What reference genome were the data aligned to?

- This is important if you need to visualize data to a specific genomic region, as different reference genomes have different maps/locations that they align to

Are the data summarized/quantified as gene expression counts, or do they only provide the raw FASTQ data?

- Raw data will need to be processed/aligned/quantified using the specifications outlined in the paper

If they provide DGE output, what was the model design? Which group is being tested for fold change? Will this be sufficient for your investigation, or will you need sample-level counts?

Most RNA-Seq data in the US is deposited in NCBI **G**ene **E**xpression **O**mnibus (GEO)

Identify the GSE number to query in NCBI GEO

Newer uploads will have more user-friendly options for collecting data

- For this example, you can use the NCBI tool to perform your own DGE analysis or download different count tables

ing interests. **Data and materials availability:** All data associated with this paper are included in the manuscript and the supplementary materials. Requests for resources and reagents should be directed to and will be fulfilled by the corresponding author, C.L.M. (cmackall@stanford.edu). RNA-, ATAC-, and ChIP-seq data have been deposited to the Gene Expression Omnibus under the accession no. GSE164950.

Weber et al. 2021. *Science*

Nearly all GEO queries will have data towards the bottom of the page

The file name suggests that these are log2 TPM counts for the data referenced in the publication

Supplementary file	Size	Download	File type/resource
GSE164949_Processed_log2TPM_Tuning-CAR-dataset.csv.gz	7.4 Mb	(ftp)(http)	CSV

SRA Run Selector

Clicking on SRA Run Selector will give us more information we might want about the samples

You can also download individual raw files (if not restricted access)

SRA Run Selector

Found 33 Items

Search within results

Clear

	Run	BioSample	AvgSpotLen	Bases	Bytes	culture_conditions	Experiment	GEO_Accession	individual	LibraryLayout	create_date	Sample Name	timepoint
1	SRR13447187	SAMN17348215	100	6.18 G	3.38 Gb	Shield-1 D1-15 + anti-PD-1 D7-15 + complete medium	SRX9860436	GSM5023691	Donor0022	PAIRED	2021-01-15 21:00:00Z	GSM5023691	Day 15
2	SRR13447188	SAMN17348214	100	6.21 G	3.39 Gb	Shield-1 D1-15 + complete medium	SRX9860437	GSM5023692	Donor0022	PAIRED	2021-01-15 22:00:00Z	GSM5023692	Day 15
3	SRR13447189	SAMN17348248	100	6.21 G	3.37 Gb	Complete medium	SRX9860438	GSM5023693	Donor0022	PAIRED	2021-01-15 21:06:00Z	GSM5023693	Day 15
4	SRR13447190	SAMN17348247	100	6.19 G	3.38 Gb	Shield-1 D1-11 + complete medium	SRX9860439	GSM5023694	Donor0614	PAIRED	2021-01-15 21:51:00Z	GSM5023694	Day 15
5	SRR13447191	SAMN17348246	100	6.21 G	3.38 Gb	Shield-1 D1-7 + complete medium	SRX9860440	GSM5023695	Donor0614	PAIRED	2021-01-15 21:06:00Z	GSM5023695	Day 15
6	SRR13447192	SAMN17348245	100	6.16 G	3.37 Gb	Shield-1 D1-15 + anti-PD-1 D7-15 + complete medium	SRX9860441	GSM5023696	Donor0614	PAIRED	2021-01-15 20:55:00Z	GSM5023696	Day 15
7	SRR13447193	SAMN17348244	100	6.28 G	3.43 Gb	Shield-1 D1-15 + complete medium	SRX9860442	GSM5023697	Donor0614	PAIRED	2021-01-15 20:54:00Z	GSM5023697	Day 15
8	SRR13447194	SAMN17348243	100	6.19 G	3.37 Gb	Complete medium	SRX9860443	GSM5023698	Donor0614	PAIRED	2021-01-15 21:55:00Z	GSM5023698	Day 15
9	SRR13447195	SAMN17348242	100	6.74 G	3.89 Gb	Complete medium	SRX9860444	GSM5023699	Donor0022	PAIRED	2021-01-15 22:24:00Z	GSM5023699	Day 7
10	SRR13447196	SAMN17348241	100	6.68 G	3.89 Gb	Shield-1 D1-7 + complete medium	SRX9860445	GSM5023700	Donor0022	PAIRED	2021-01-15 21:04:00Z	GSM5023700	Day 7
11	SRR13447197	SAMN17348240	100	6.92 G	4.02 Gb	Complete medium	SRX9860446	GSM5023701	Donor0022	PAIRED	2021-01-15 21:31:00Z	GSM5023701	Day 11
12	SRR13447198	SAMN17348239	100	7.18 G	4.16 Gb	Shield-1 D1-11 + complete medium	SRX9860447	GSM5023702	Donor0022	PAIRED	2021-01-15 21:04:00Z	GSM5023702	Day 11
13	SRR13447199	SAMN17348238	100	6.92 G	4.02 Gb	Shield-1 D1-11 + anti-PD-1 D7-11 + complete medium	SRX9860448	GSM5023703	Donor0022	PAIRED	2021-01-15 21:11:00Z	GSM5023703	Day 11
14	SRR13447200	SAMN17348237	100	6.69 G	3.90 Gb	Shield-1 D1-D7 + complete medium	SRX9860449	GSM5023704	Donor0022	PAIRED	2021-01-15 21:06:00Z	GSM5023704	Day 11
15	SRR13447201	SAMN17348236	100	6.69 G	3.89 Gb	Complete medium	SRX9860450	GSM5023705	Donor0614	PAIRED	2021-01-15 21:11:00Z	GSM5023705	Day 7
16	SRR13447202	SAMN17348235	100	6.93 G	4.03 Gb	Shield-1 D1-7 + complete medium	SRX9860451	GSM5023706	Donor0614	PAIRED	2021-01-15 21:08:00Z	GSM5023706	Day 7

Count table options are sometimes available through NCBI

If we don't have a need for transformed log2 TPM counts or want to perform our own normalization for the counts, newer options allow for an NCBI raw count matrix or annotated counts

Information about GEO data organization is provided in the GEO overview. GEO FTP site structure and available formats are described in README.txt. Additional download options are described at Download GEO data.

Submitter-supplied data

Series SOFT file
GSE164949_family.soft.gz — 4.8 Kb

Series MINiML file
GSE164949_family.xml.tgz — 5.1 Kb

Series Matrix file
GSE164949_series_matrix.txt.gz — 4.4 Kb

Series supplementary data
GSE164949_Processed_log2TPM_Tuning-CAR-dataset.csv.gz — 7.4 Mb

NCBI-generated data BETA

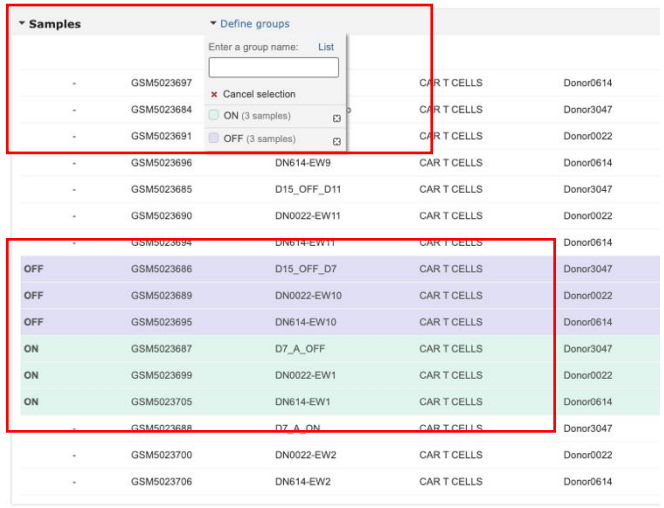
Series RNA-seq raw counts matrix
Raw count matrices may be suitable for input into differential expression analysis tools.
GSE164949_raw_counts_GRCh38.p13_NCBI.tsv.gz — 1.4 Mb — 2022-12-17

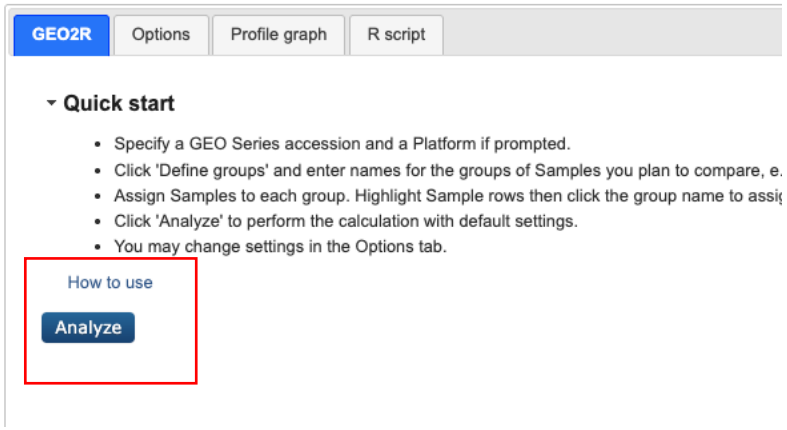
Series RNA-seq normalized counts matrix
Normalized count matrices may be suitable for analyzing and visualizing gene expression abundance.
GSE164949_norm_counts_FPKM_GRCh38.p13_NCBI.tsv.gz — 2.1 Mb — 2022-12-17
GSE164949_norm_counts_TPM_GRCh38.p13_NCBI.tsv.gz — 2.1 Mb — 2022-12-17

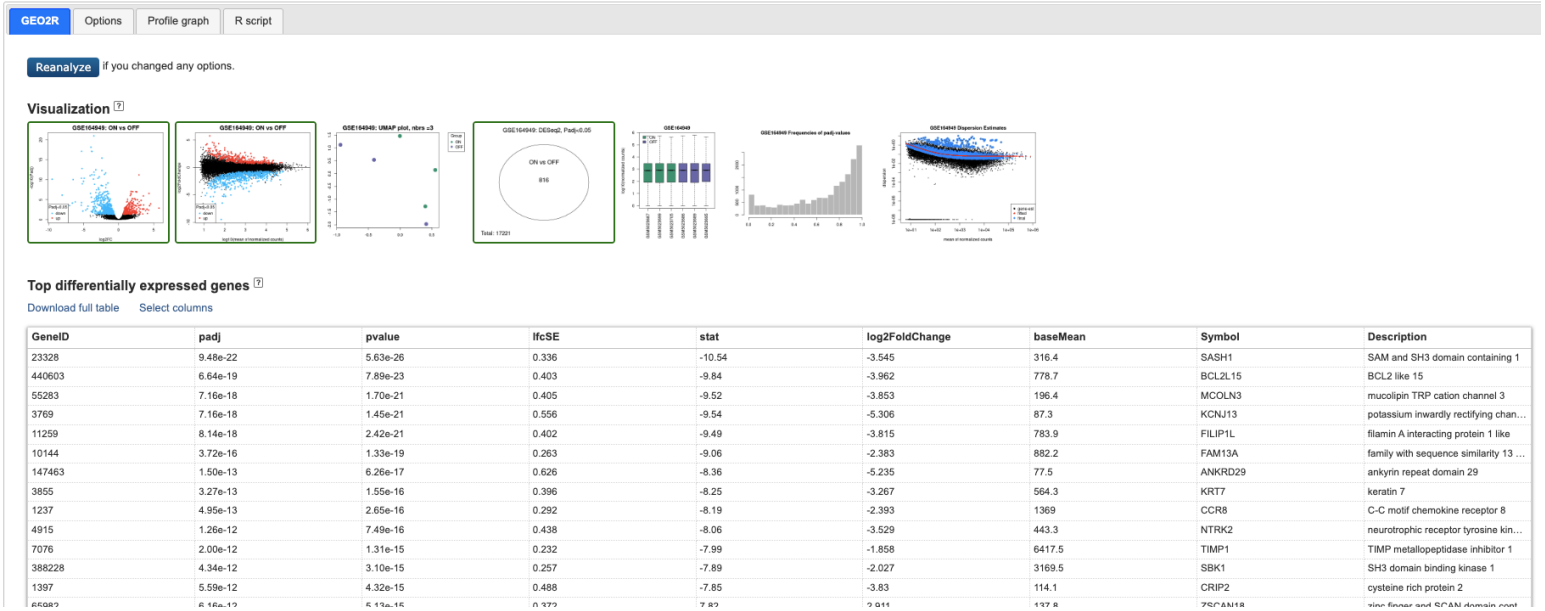
Human gene annotation table
Gene annotation information is intended to be used in conjunction with raw and normalized count matrices.
The *Homo sapiens* Annotation Release 109.20190905 was used for gene annotation.
Human.GRCh38.p13.annot.tsv.gz — 4.2 Mb — 2022-11-08

DGE can sometimes be performed through NCBI directly

If the “Analyze with GEO2R” option is available, you can select which samples to test as well as generate figures







GeneID	padj	pvalue	lfcSE	stat	log2FoldChange	baseMean	Symbol	Description
23328	9.48e-22	5.63e-26	0.336	-10.54	-3.545	316.4	SASH1	SAM and SH3 domain containing 1
440603	6.64e-19	7.89e-23	0.403	-9.84	-3.962	778.7	BCL2L15	BCL2 like 15
55283	7.16e-18	1.70e-21	0.405	-9.52	-3.853	196.4	MCOLN3	mucolipin TRP cation channel 3
3769	7.16e-18	1.45e-21	0.556	-9.54	-5.306	87.3	KCNJ13	potassium inwardly rectifying chan…
11259	8.14e-18	2.42e-21	0.402	-9.49	-3.815	783.9	FILIP1L	filamin A interacting protein 1 like
10144	3.72e-16	1.33e-19	0.263	-9.06	-2.383	882.2	FAM13A	family with sequence similarity 13 …
147463	1.50e-13	6.26e-17	0.626	-8.36	-5.235	77.5	ANKRD29	ankyrin repeat domain 29
3855	3.27e-13	1.55e-16	0.396	-8.25	-3.267	564.3	KRT7	keratin 7
1237	4.95e-13	2.65e-16	0.292	-8.19	-2.393	1369	CCR8	C-C motif chemokine receptor 8
4915	1.26e-12	7.49e-16	0.438	-8.06	-3.529	443.3	NTRK2	neurotrophic receptor tyrosine kin…
7076	2.00e-12	1.31e-15	0.232	-7.99	-1.858	6417.5	TIMP1	TIMP metallopeptidase inhibitor 1
388228	4.34e-12	3.10e-15	0.257	-7.89	-2.027	3169.5	SBK1	SH3 domain binding kinase 1
1397	5.59e-12	4.32e-15	0.488	-7.85	-3.83	114.1	CRIP2	cysteine rich protein 2

Other RNA-Seq databases are available as well

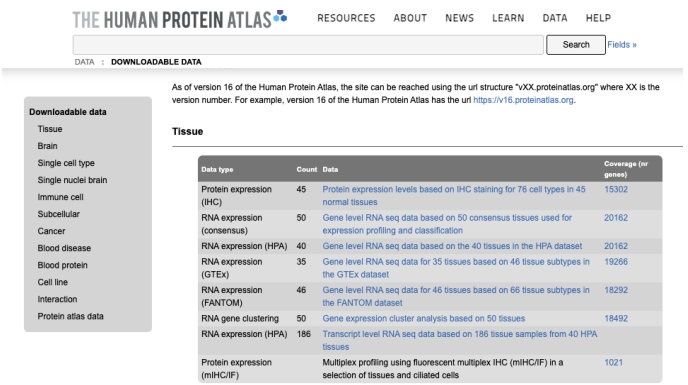
Another common repository for RNA-Seq datasets (primarily in the UK) is EMBL-EBI ArrayExpress

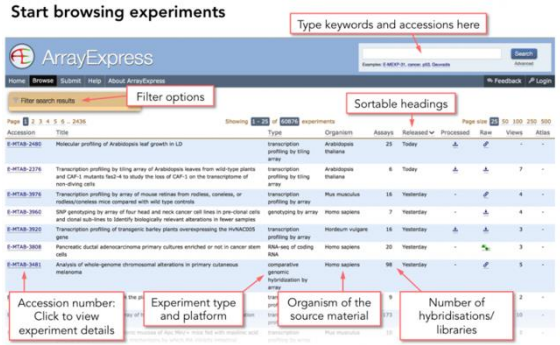
- These usually have E-MTAB numbers associated with them

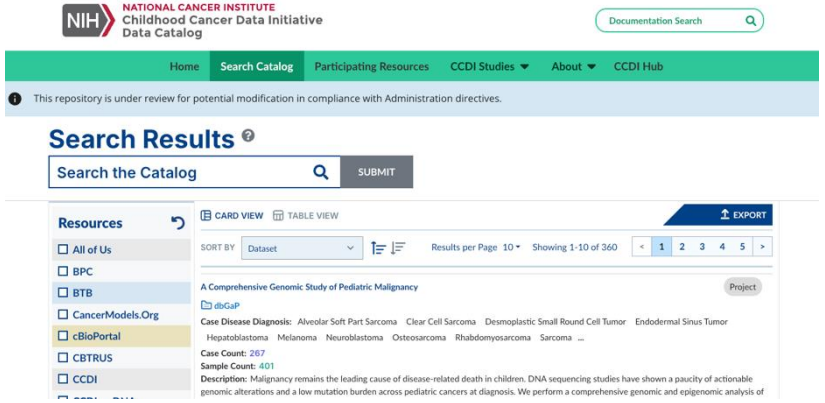
Human Protein Atlas has many different downloadable bulk or single-cell NGS datasets for download

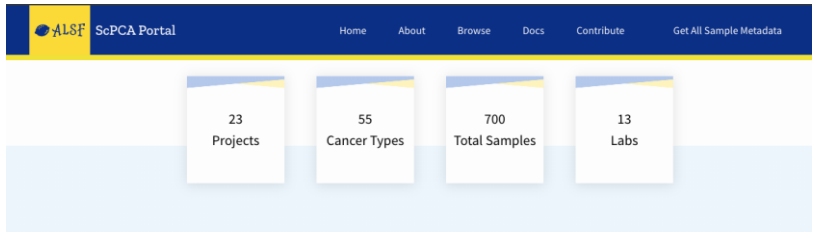
NCI has a pediatric cancer data catalog with a variety of different sample types

Foundations like ALSF, St. Jude, and MD Anderson have their own repositories as well









Outline

Background

- Affymetrix platform → bulk RNA-Seq → scRNA-Seq
- Data types to consider
- Publicly available databases
  - What to look for when re-analyzing data from a publication

Statistical Methodologies

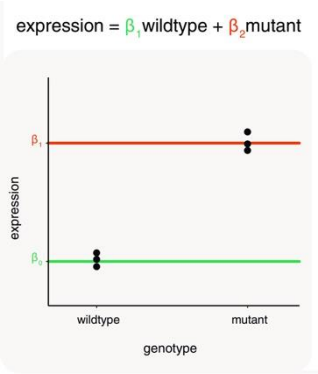
- Model considerations
- Software assumptions

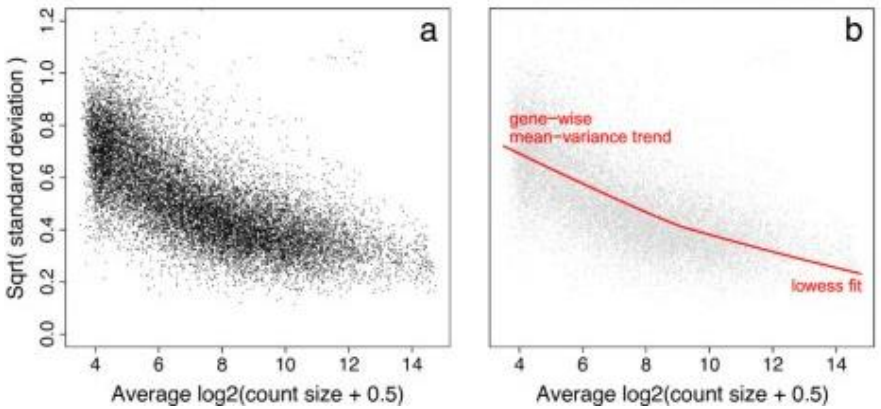
Bioinformatics tools

- Shell/R scripting best practices
- Processing raw FASTQ files to BAM files
- Quantifying BAM files
- Normalizing/Transforming data and statistical Analysis (Hands-on portion)

Two common DGE tools commonly used have slightly different model assumptions

Differential Analysis asks whether or not there's a significant difference in expression of our treatment group vs the control group. Individual genes are usually tested under this model design:





Limma/Voom or edgeR was originally developed for Microarray analysis but has been adapted for RNA-Seq. It normalizes/transforms the counts and uses gene-weighted mean-variance measures to test differences. The transformation and testing allow for complex model designs, and model assumptions and interpretations are similar to traditional regression models.

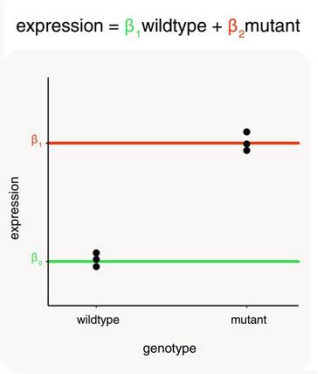
- Best used when there are large discrepancies in sequencing library size (about threefold ratio difference)
- Must filter out low-expressing genes before performing analysis
- Usually viewed as providing more conservative model estimates

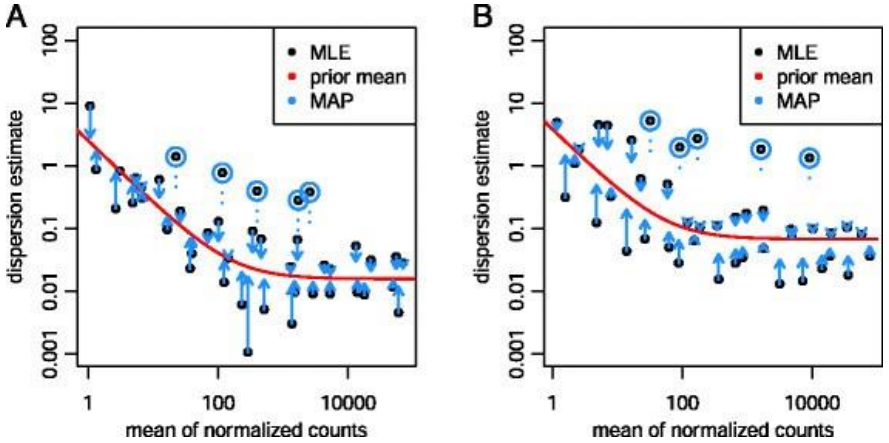
Figure 1. Law et al. 2020. *F1000 Research*

Figure 2. Law et al. 2014. *Genome Biology*

Two common DGE tools commonly used have slightly different model assumptions

Differential Analysis asks whether or not there's a significant difference in expression of our treatment group vs the control group. Individual genes are usually tested under this model design:





DESeq2 uses a Negative-Binomial distribution to calculate changes in gene expression. This framework is typically used when using "count" data and uses shrinkage estimation to give more stable fold-changes.

- More widely used in human studies
- Suggested not to filter out lowly expressed genes
- Not considered as conservative as limma/voom

Figure 1. Law et al. 2020. *F1000 Research*

Figure 1. Love et al. 2014. *Genome Biology*

Outline

Background

- Affymetrix platform → bulk RNA-Seq → scRNA-Seq
  - FASTA and FASTQ file format
  - BAM file format and count quantification
- Data types to consider
- Publicly available databases
  - What to look for when re-analyzing data from a publication

Statistical Methodologies

- Model considerations
- Software assumptions

Bioinformatics tools

- Shell/R scripting best practices
- DGE Statistical Analysis using R (Hands-on portion)

What is scripting and why do we use it?

Consider scripting like your lab notebook R and Python have tools to help with documentation.

- R Studio
- Jupyter Notebook/Spyder
- Microsoft Visual Studio

”Shell” scripting helps you keep track of commands you use that you can implement in Linux/Unix environment.

- You can use tools to remotely access a server and run commands
  - MobaXterm
  - PuTTY
  - Terminal
- Most people have a preferred editor to use to create and save these commands
  - Notepad ++
  - Phoenix Code

Things to consider when documenting code

Be sure to use “commenting” whenever possible

- Usually, when people use “#” or ”/* */” around a code block, it helps explain what that code is supposed to do

- When performing a set of manipulations, it’s nice to see what is being done so you can address that specific code block or code lines when troubleshooting

- Try to document even when it may seem unnecessary or too simplistic to do so.
  - When you come back to a script several weeks or months later, it will help you keep track of the logic you were using at the time
  - If you have to share your script with someone else, you never know at what level their coding is for that specific language or toolset

Resources for when nothing is working

When encountering errors for a long period of time, the best thing to do is walk away for a bit

- Physically remove yourself from the computer, take a break, walk around, literally do anything else

After taking a break, here are some helpful resources (Google is your friend)

- Search for the relevant part of your error message, then the software/language you're using
  - Stack overflow is a really good resource. It's very likely that others have encountered your issue and have already asked this forum about it. Users will upvote the best response.

  - Biostars is a forum that's more specifically bioinformatics-focused and sometimes answers more questions about practical considerations for study design or theoretical rationale

AI can be helpful but **use caution**

- Be sure to use a specific prompt about what you'd like to accomplish. Tell it what language you're using, what you're starting with, and what you want it to give you. Use caution before running; make sure you're not overwriting important data in the process.
- Make sure it actually gave you what you asked for. You will probably have to ask it to refine portions of code.
- Never provide it with PHI

SCRI Research Scientific Computing has a TON Resources, take advantage!

Everyone should have access to the Bioinformatics & Scientific Computing page

- There are a variety of different in-depth courses with videos and presentations at your fingertips, some of which include:
  - Introduction to R for Biologists
  - Introduction to Python for Biologists
  - Bulk RNA-Seq analysis in Linux and R
  - Single cell analysis in Linux and R

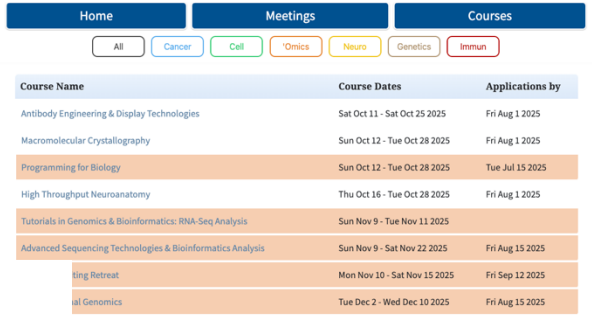
- Also the HPC page has an excellent user guide on how to use Sasquatch

- You also may know that they hold office hours on-site and virtually every Tuesday at 11 AM

Recommended Workshops

Cold Spring Harbor

- I haven't personally taken any of these but hear good things from others



Home | Meetings | Courses

All | Cancer | Cell | 'Omics | Neuro | Genetics | Immun

Course Name	Course Dates	Applications by
Antibody Engineering & Display Technologies	Sat Oct 11 - Sat Oct 25 2025	Fri Aug 1 2025
Macromolecular Crystallography	Sun Oct 12 - Tue Oct 28 2025	Fri Aug 1 2025
Programming for Biology	Sun Oct 12 - Tue Oct 28 2025	Tue Jul 15 2025
High Throughput Neuroanatomy	Thu Oct 16 - Tue Oct 28 2025	Fri Aug 1 2025
Tutorials in Genomics & Bioinformatics: RNA-Seq Analysis	Sun Nov 9 - Tue Nov 11 2025	
Advanced Sequencing Technologies & Bioinformatics Analysis	Sun Nov 9 - Sat Nov 22 2025	Fri Aug 15 2025
ting Retreat	Mon Nov 10 - Sat Nov 15 2025	Fri Sep 12 2025
al Genomics	Tue Dec 2 - Wed Dec 10 2025	Fri Aug 15 2025

University of Washington Summer Short Courses

- A little more focused on statistical theory
- Exercises can be a bit clunky or difficult to execute, depending on who is teaching

Institutes & Dates

Summer Institute in Statistics for Clinical and Epidemiological Research (SISCER)

July 8 - July 31, 2025

Live virtual sessions will be held 8:30 a.m. to noon PDT (11:30 a.m. – 3 p.m. EDT) unless noted otherwise

- Design and Analysis of Clinical Trials, Tuesday, July 8 – Thursday, July 10
- Analyzing Data from Complex Surveys, Tuesday, July 8 – Wednesday, July 9 (afternoons 1 p.m. – 4:30 p.m. PDT; 4-7:30 p.m. EDT)
- Modern Statistical Learning for Observational Data, Tuesday, July 8 – Friday, July 11
- Introduction to Survival Analysis, Thursday, July 10 – Friday, July 11 (afternoons 1 p.m. – 4:30 p.m. PDT; 4 - 7:30 p.m. EDT)
- Special Topics in Clinical Trials: Interpreting Exploratory Analysis & When Are Biomarkers Valid Surrogates?, Friday, July 11
- Improving Precision and Power in Randomized Trials by Leveraging Baseline Variables, Monday, July 14 – Tuesday, July 15
- Generalized Estimating Equations and Mixed-Effects Models for Longitudinal Data Analysis, Monday, July 14 – Wednesday, July 16
- Design and Analysis of Two-Phase Studies, Wednesday, July 16 – Thursday, July 17 (afternoons 1 p.m. – 4:30 p.m. PDT; 4 - 7:30 p.m. EDT)
- Joint Modeling of Longitudinal and Survival Data, Thursday, July 17 – Friday, July 18
- Evaluation of Biomarkers and Risk Models, Monday, July 21 – Tuesday, July 22
- Using Causal Graphs in Epidemiological Research, Tuesday, July 22
- Absolute Risk: Methods and Applications in Clinical Care and Public Health, Wednesday, July 23 – Thursday, July 24
- Causal Inference with Observational Data: Common Designs and Statistical Methods, Wednesday, July 23 – Friday, July 25
- Missing Data Methods, Monday, July 28 - Tuesday, July 29
- Irregular Assessment Times in Randomized Controlled Trials, Wednesday July 30 - Thursday, July 31
- Propensity Scores, Wednesday, July 30 - Thursday, July 31

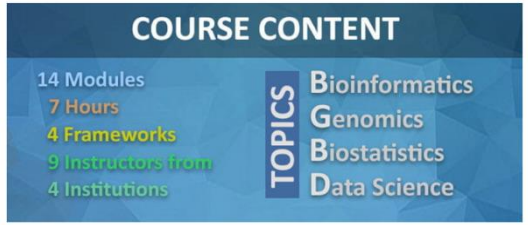
Summer Institute in Statistics for Big Data (SISBID)

July 21 – August 1, 2025

- Data Wrangling with R, Monday, July 21 – Wednesday, July 23 (morning)
- Data Visualization , Wednesday, July 23 (afternoon) – Friday, July 25
- Supervised Methods for Statistical Machine Learning, Monday, July 28 – Wednesday, July 30 (morning)
- Unsupervised Methods for Statistical Machine Learning, Wednesday, July 20 (afternoon) – Friday, August 1

University of Alabama at Birmingham

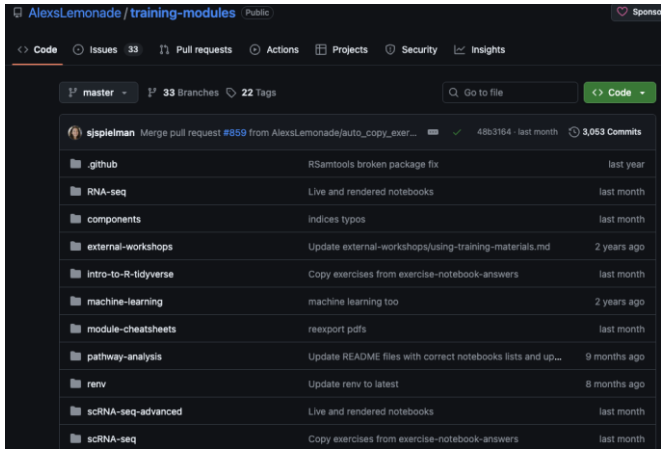
- Used to have an organized/structured course
- YouTube series about a variety of –omics tutorials



- Computing & Informatics
- Next Generation Sequencing Analysis
- GWAS Methods & Analysis
- Somatic Analysis

Alex Lemonade Stand Foundation

- Probably my favorite hands-on workshop I've taken
- Very well-documented GitHub repository for a variety of bioinformatics tutorials



AlexsLemonade / training-modules (Public)

Code | Issues 33 | Pull requests | Actions | Projects | Security | Insights

master | 33 Branches | 22 Tags | Go to file | Code

sjspielman Merge pull request #859 from AlexsLemonade/auto_copy_exer... 48b3164 · last month 3,053 Commits

.github	RSamtools broken package fix	last year
RNA-seq	Live and rendered notebooks	last month
components	indices typos	last month
external-workshops	Update external-workshops/using-training-materials.md	2 years ago
intro-to-R-tidyverse	Copy exercises from exercise-notebook-answers	last month
machine-learning	machine learning too	2 years ago
module-cheatsheets	reexport pdfs	last month
pathway-analysis	Update README files with correct notebooks lists and up...	9 months ago
renv	Update renv to latest	8 months ago
scRNA-seq-advanced	Live and rendered notebooks	last month
scRNA-seq	Copy exercises from exercise-notebook-answers	last month